

Learning Objectives

- Train and evaluate a Decision Tree Regressor.
- Train and evaluate a Random Forest Regressor.
- Explain bias, variance, and the bias–variance tradeoff in plain language.
- Empirically demonstrate, using code and plots, how ensembling (bagging) reduces variance.

Dataset

Use the following noisy non-linear function so your fit can be visualized easily:

```
import numpy as np
from sklearn.model_selection import train_test_split

np.random.seed(42)

def true_function(x):
    return np.sin(1.5 * np.pi * x)

n_samples = 200
X = np.sort(np.random.rand(n_samples))
y = true_function(X) + np.random.normal(0, 0.25, n_samples)
X = X.reshape(-1, 1)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

(Alternative: you may use `sklearn.datasets.fetch_california_housing()` for a real-world tabular dataset instead of the toy sine wave, if your instructor permits — the tasks below apply the same way.)

Part A — Decision Tree Regressor

1. Train a single `DecisionTreeRegressor` with `max_depth=None` (fully grown).
2. Compute Train MSE and Test MSE.
3. Repeat for `max_depth = 1, 2, 3, 5, 8, 12, None`. Plot Train MSE and Test MSE vs `max_depth` on the same chart.
4. In your own words: At `max_depth=1`, is the model biased or high-variance? Why?
5. In your own words: At `max_depth=None`, what happens to Train MSE vs Test MSE? What does this indicate?

Part B — Random Forest Regressor

6. Train a `RandomForestRegressor` with `n_estimators=200`, `max_depth=None`.
7. Compute Train MSE and Test MSE, and compare with the single fully-grown tree from Part A.
8. Vary `n_estimators = 1, 5, 10, 20, 50, 100, 200` and plot Train/Test MSE vs `n_estimators`.
9. Why does Test MSE improve/plateau as `n_estimators` increases, while Train MSE stays roughly flat?
10. Does adding more trees ever cause the Random Forest to overfit more? Justify your answer using the concept of averaging.

Part C — Comparative Study (Bias–Variance)

11. Train 30 individual, fully-grown Decision Trees, each on a different bootstrap sample of the training data. Overlay all 30 predictions on one chart.
12. On a separate chart, plot the prediction of a single Random Forest (200 trees) trained on the original training data.
13. Compare the two charts: which shows more spread/disagreement (variance)? Which is closer to the true function on average (bias)?
14. Write one paragraph explaining why Random Forest reduces variance compared to a single Decision Tree, referencing bagging and averaging of largely uncorrelated errors.

Submission Instructions

- Submit a single Jupyter Notebook (.ipynb) containing: all code, all required plots (minimum 3), and written answers to every question in Parts A, B, and C zip in google form